Gaussian Splatting and NeRF
Worked solutions

Computer Vision 2026
University of Bern · Computer Vision Group

Interactive lab: sosekie.github.io/Unibe-CVG-CV-Interactive-Labs/gaussian-splatting-nerf/
Source code: github.com/Sosekie/Unibe-CVG-CV-Interactive-Labs

Learning objectives

- Connect Gaussian shape to covariance eigenvectors, variances, and normalization.
- Distinguish coefficient fitting, geometry fitting, and image-based inverse rendering.
- Explain projection, visibility, optical thickness, and pixel gradients.
- Compare Gaussian primitives with neural fields and design controlled evaluation.

Key concepts: identifiability, conditioning, covariance, gradient flow, compositing, multi-view geometry, density control, ray integration, and generalization.

Section	Experiment	Questions
1	Approximate a function in 1D	1.1 and 1.2
2	Covariance and a 2D input domain	2.1 and 2.2
3	Fit a 3D scalar function	3.1 and 3.2
4	Project, composite, and expose ambiguity	4.1 and 4.2
5	Recover a scene from images	5.1 and 5.2
6	NeRF: a neural field along camera rays	6.1 and 6.2
7	Transfer to the real captured Garden	7.1 and 7.2

Working with the tutorial

Compare each worked derivation with your own reasoning, then test the prediction in the lab. Distinguish evidence for one experiment from a general guarantee.

This handout follows the reviewed interactive tutorial. Lecture-slide references and a numbered course worksheet assignment remain pending. The NeRF experiment trains a small 2D cross-section; the final Garden renders a pretrained Gaussian Splatting model.

> Local use: serve the lab's **dist/** directory with Python 3. Sections 1–6 use no remote assets. The optional Garden requires network access and WebGL and streams approximately 187 MB.

References: Kerbl et al., 3D Gaussian Splatting (2023); Mildenhall et al., NeRF (2020). See the lab README for conventions and teaching limits.

1.1 · IDENTIFIABILITY · LINEAR FITTING

Interactive experiment · Source code

You fit two 1D Gaussian basis functions with exactly the same mean and width. Only their amplitudes are free. The target samples are fixed.

What can least squares determine about the two amplitudes?

> **Answer B.** Their sum can be determined, but the individual amplitudes are not identifiable.

The two design-matrix columns are identical: $\Phi = [\varphi\ \varphi]$. Every prediction depends on $a_1 + a_2$. Adding δ to one amplitude and subtracting δ from the other changes neither the prediction nor its loss.

There are infinitely many coefficient pairs at the optimum. A numerical solver can return one representative, but that does not make the parameters identifiable. More samples cannot distinguish two exactly identical columns. Nearly identical kernels instead make the coefficients sensitive to small perturbations.

$$\hat{y} = (a_1 + a_2)\phi, \qquad \operatorname{rank}(\Phi) = 1, \qquad \Phi \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 0$$

Follow-up

Would adding a penalty $\lambda\|a\|^2$ make the amplitudes identifiable from the observations?

A positive quadratic penalty selects a unique minimum-norm pair (here equal amplitudes), but this uniqueness comes from the added preference. The observations still constrain only the sum.

Test it in the lab

Set N = 2. Give both kernels the same mean and width, then fit amplitudes. Inspect the reported basis rank. Move one mean slightly and compare rank and coefficients.

> *A good function fit need not identify its individual Gaussian components.*

1.2 · MODEL ASSUMPTIONS · SIGNED WEIGHTS

Interactive experiment · Source code

After an unconstrained amplitude fit, a nonnegative target is approximated using both positive and negative Gaussian coefficients. Small regions of the approximation fall below zero.

Which interpretation is consistent with the model used in this lab?

> **Answer C.** Signed coefficients are valid for function approximation; nonnegativity would require a different constrained fit.

Here φ_i is an unnormalized basis function and $\hat{y} = \Sigma_i a_i \varphi_i$ is a scalar function. Unconstrained least squares allows cancellation: a broad positive contribution can be corrected by a narrower negative contribution. Neither nonnegative predictions nor coefficients summing to one are required.

A probability mixture instead uses normalized component densities and nonnegative weights summing to one. Nonnegative least squares would impose $a_i \geq 0$ and change the feasible predictions. Its optimum cannot have a smaller sample SSE than the unconstrained optimum using the same basis.

$$\text{Unconstrained: } \min_{a} \|\Phi a - y\|^2$$
$$\text{Nonnegative: } \min_{a_i \geq 0} \|\Phi a - y\|^2$$

Follow-up

Why can a low sample error coexist with large, unstable positive and negative coefficients?

Nearly dependent basis columns allow large contributions to cancel. The sampled sum can remain accurate while individual coefficients are sensitive to perturbations. Rank, conditioning, off-sample behavior, and coefficient size tell us more than training error alone.

Test it in the lab

Use the plateau + sharp-detail target with several kernels and fit amplitudes. Inspect individual contributions and the residual. If a coefficient is negative, set it to zero manually and compare sample error before refitting.

> *Function weights, probability-mixture weights, and rendering opacities have different meanings and constraints.*

2.1 · COVARIANCE · PRINCIPAL AXES

Interactive experiment · Source code

A 2D Gaussian has Σ = [[0.20, 0.16], [0.16, 0.20]]. The diagonal entries are equal.

What shape does this covariance describe?

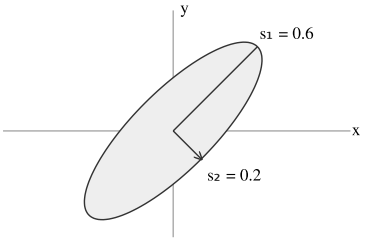
> **Answer A.** An elongated Gaussian, with standard deviations 0.6 and 0.2 along axes rotated by 45°.

Eigenvalues solve the characteristic equation det(Σ − λI) = 0. For a symmetric 2×2 matrix Σ = [[u,v],[v,w]], expand the determinant to obtain λ² − (u+w)λ + uw − v² = 0. The quadratic formula gives λ± = (u+w ± √((u−w)² + 4v²))/2.

Here u = w = 0.20 and v = 0.16. With equal diagonal entries d, the eigenvalues simplify to d+v and d−v, associated with directions (1,1) and (1,−1). Thus λ = 0.36 and 0.04. Taking square roots gives principal standard deviations 0.6 and 0.2. Unit eigenvectors are (1,1)/√2 and (1,−1)/√2, so the long axis is at 45°.

Equal marginal variances do not imply rotational symmetry. Isotropy requires Σ = σ²I, including zero off-diagonal entries. Conversely, an isotropic Gaussian is unchanged by rotation: its angle cannot be recovered from its function values.

$$
\begin{aligned}
\det(\Sigma - \lambda I) &= (u-\lambda)(w-\lambda) - v^2 = 0 \\
\lambda_{\pm} &= \frac{u + w \pm \sqrt{(u-w)^2 + 4v^2}}{2} \\
&= \frac{0.40 \pm 0.32}{2} = 0.36,\ 0.04 \\
s_i &= \sqrt{\lambda_i} = 0.6,\ 0.2
\end{aligned}
$$



The 1σ ellipse: axes at ±45°, with semiaxes 0.6 and 0.2.

Follow-up

What happens to the usefulness of a rotation gradient as the two principal scales become equal?

The function becomes independent of angle at exact isotropy, so the angle derivative vanishes. Near isotropy, orientation has little effect and can be weakly constrained. A rotation parameter is not always an observable property.

Test it in the lab

Select one 2D Gaussian, set scales to 0.60 and 0.20 and rotation to 45°, then inspect its covariance. Make it isotropic and rotate again; predict which visual and numerical quantities should stay fixed.

> *Read Gaussian shape from covariance eigenpairs, and distinguish parameters from observable effects.*

2.2 · GAUSSIAN BEHAVIOR · NORMALIZATION

Interactive experiment · Source code

Consider one unnormalized 2D kernel a exp(−½(x−μ)ᵀΣ⁻¹(x−μ)). Double both principal scales while keeping its amplitude a fixed.

What happens to its peak value and its integral over the whole plane?

> **Answer B.** The peak stays fixed and the integral becomes four times as large.

The principal scales have a direct PCA interpretation. For samples from one normalized Gaussian N(μ,Σ), PCA of the centred samples estimates the eigendecomposition $\Sigma = U\,\mathrm{diag}(\lambda_i)\,U^\top$. The columns of U are the principal directions, λ_i are the variances along them, and the Gaussian scales $s_i = \sqrt{\lambda_i}$ are the corresponding standard deviations. Finite samples give estimates; eigenvector signs are arbitrary, and repeated eigenvalues do not define unique directions within their eigenspace.

At the mean, the exponent is zero, so the value remains a. The integral of this unnormalized 2D kernel is $a \cdot 2\pi \cdot \sqrt{\det(\Sigma)}$. Doubling both scales multiplies each covariance eigenvalue by four, and $\sqrt{\det(\Sigma)}$ by four.

A normalized probability density includes the inverse determinant factor and would instead keep unit integral while lowering the peak. The lab does not include that normalization. A larger Gaussian therefore changes how much of the domain it covers, even when its peak is unchanged.

This PCA interpretation applies to each Gaussian component. PCA of a complete probability mixture also reflects variation between component means, so its principal axes can differ from those of the individual Gaussians. The signed function expansion in this lab is not automatically a probability distribution.

$$\Sigma = U\,\mathrm{diag}(\lambda_i)U^\top, \quad s_i = \sqrt{\lambda_i}$$

$$\phi(\mu) = 1, \quad \int a\phi(x)\,dx = a(2\pi)^{d/2}\sqrt{\det\Sigma}$$

$$s_i \mapsto k s_i \quad \Longrightarrow \quad \text{integral multiplied by } k^d$$

Follow-up

Does the fourfold integral increase imply fourfold image brightness after projecting and alpha-compositing a 3D splat?

No. The scalar-field integral is not the image-formation model. Projection, camera pose, opacity saturation, overlap, and transmittance determine pixel contributions. The peak opacity of an unnormalized splat stays fixed at its projected mean.

Test it in the lab

In 2D, isolate a contribution by setting other amplitudes to zero. Change scales from 0.25 to 0.50 while keeping amplitude fixed. Compare peak value, footprint, and residual; account for the finite plotted domain.

> *Scaling a Gaussian changes coverage; its peak behavior depends on whether the kernel is normalized.*

3.1 · OPTIMIZATION · GRADIENT FLOW

Interactive experiment · Source code

The 3D model is ŷ(x) = aφ(x;μ,Σ), fitted by mean squared error to sampled target values. Consider the raw gradient at a = 0, with positive scales.

Which parameters can receive a nonzero loss gradient at that instant?

> **Answer C.** Amplitude can receive a nonzero gradient, while the mean and covariance gradients are zero.

The derivative with respect to amplitude is φ. The mean derivative is aφΣ⁻¹(x−μ); covariance and scale derivatives also contain a. At a = 0, the current loss is locally insensitive to geometry, even if prediction error is large. The amplitude derivative can remain nonzero.

Once amplitude changes, geometry gradients can become nonzero. Initialization matters in another way: if a narrow kernel lies far from all informative samples, φ is tiny there and even the amplitude gradient can be weak. Large residuals alone do not guarantee useful parameter gradients.

$$r_j = \hat{y}(x_j) - f(x_j), \quad L = \frac{1}{M}\sum_j r_j^2$$

$$\frac{\partial L}{\partial a} = \frac{2}{M}\sum_j r_j \phi_j$$

$$\nabla_\mu L = \frac{2}{M}\sum_j r_j a \phi_j \Sigma^{-1}(x_j - \mu)$$

Follow-up

Can Adam momentum move a mean when its current raw gradient is zero?

Yes, if its stored first moment is nonzero from previous updates. A zero current gradient is not the same as a zero optimizer update. For the proposed lab experiment, editing an amplitude clears optimizer moments, so the first update isolates the raw-gradient effect.

Test it in the lab

Choose the Gaussian-bump target and N = 1. Set amplitude to exactly zero; note the mean and scales. Apply one update and inspect what changed. Apply more updates, then compare with a narrow kernel placed far from the target.

> *To diagnose fitting, reason about the Jacobian of the prediction as well as the residual.*

3.2 · APPROXIMATION · CAPACITY VS OPTIMIZATION

Interactive experiment · Source code

One Gaussian fits the ring-shaped 3D target poorly after many updates. Its slices fill the hole or miss much of the ring.

What can be concluded from the Gaussian model itself, without assuming the optimizer found a global minimum?

> **Answer A.** One Gaussian cannot represent the target exactly; more optimization may help, but cannot remove this representation mismatch.

For a positive amplitude, every nonempty positive level set of a Gaussian is an ellipsoid enclosing one maximum. Changing mean, covariance, or amplitude cannot create the ring's hole. A nonpositive amplitude cannot reproduce its positive values either. Exact representation is therefore unavailable with one kernel.

This does not tell us whether the current parameters are optimal within that restricted family. There may be both approximation error and optimization error. Several kernels can cover different parts of the ring, but their nonlinear fit can still depend on initialization and the update budget.

$$a\phi(x) = \tau > 0 \quad \Longrightarrow \quad (x-\mu)^\top \Sigma^{-1}(x-\mu) = 2\log(a/\tau)$$

Follow-up

If the displayed volume RMSE is small, could a visually important part of the ring still be poorly fitted?

Yes. Uniform volume MSE averages over many nearly empty samples. A localized structure can contribute little to the global average while being badly approximated. Check multiple slices, residual location, and sampling density; consider a region-weighted diagnostic before declaring success.

Test it in the lab

Fit the ring with N = 1 and N = 8 using comparable update budgets. Compare volume RMSE and all three slices. Separate evidence of increased capacity from evidence that a particular run reached a better solution.

> *A loss plateau does not by itself diagnose a local minimum; model capacity and the measurement distribution also matter.*

4.1 · VISIBILITY · PHOTOMETRIC GRADIENTS

Interactive experiment · Source code

At one pixel, a foreground splat has $\alpha_1 = 0.98$ and a rear splat has $\alpha_2 = 0.6$. Hold geometry and opacity fixed while differentiating the pixel colour with respect to the rear RGB.

What is $\partial C/\partial c_2$ for each RGB channel, and what does it imply for fitting?

> **Answer B.** 0.012: foreground transmittance strongly attenuates the rear colour gradient at this pixel.

The foreground leaves $T_2 = 1 - 0.98 = 0.02$. The rear contribution is $T_2\alpha_2 c_2$, so the per-channel derivative is $0.02 \times 0.6 = 0.012$. A photometric loss gradient contains this factor multiplied by its pixel residual derivative.

A large pixel residual can therefore give little signal to a nearly hidden splat. With α_1 exactly one, this colour gradient would be zero at this pixel. Other pixels and camera views may expose the rear splat. Visibility influences which parameters can actually learn from each observation.

$$C = \alpha_1 c_1 + (1-\alpha_1)\alpha_2 c_2 + (1-\alpha_1)(1-\alpha_2)c_{\text{bg}}$$
$$\frac{\partial C}{\partial c_2} = (1-\alpha_1)\alpha_2 I = 0.02 \cdot 0.6\, I = 0.012\, I$$

Follow-up

Would directly adding the two coloured kernels give the same learning behavior?

No. In an additive colour sum, the rear colour derivative would lack the foreground transmittance factor. Alpha compositing couples visibility and parameter influence; it is not just the scalar basis expansion with RGB outputs.

Test it in the lab

Use three-splat overlap and sample an overlap pixel. Raise the nearest splat's opacity and track the rear weights in the table. Orbit to another view. The displayed alpha includes spatial falloff, so use the measured alpha values rather than just the opacity sliders.

> *Image fitting learns through visibility, not equally from every Gaussian at every pixel.*

4.2 · MULTI-VIEW GEOMETRY · AMBIGUITY

Interactive experiment · Source code

For an isolated Gaussian and a fixed pinhole camera, multiply its camera-relative mean and all spatial scales by k > 0. Keep RGB and opacity fixed, and keep it in front of the near plane. Its locally projected footprint is unchanged.

Which extra observation generally constrains this depth–scale ambiguity?

> **Answer C.** A calibrated image from a different camera centre, with parallax and visibility of the Gaussian.

Scaling camera-relative position preserves x/z and y/z, hence the projected mean. The pinhole Jacobian scales by 1/k while 3D covariance scales by k², preserving the locally projected covariance. The fixed pixel filter is also unchanged. The single image cannot distinguish this family of 3D explanations.

Pure rotation about the same optical centre keeps this radial scaling ambiguity. A translated camera supplies another ray origin and generally makes the two candidate means project differently. Useful baseline and visibility matter; a second image is not an unconditional guarantee of accurate or unique reconstruction.

$$\mu_{\text{new}} = e + k(\mu - e), \quad \Sigma_{\text{new}} = k^2\Sigma, \quad J_{\text{new}} = J/k$$
$$J_{\text{new}} R^c \Sigma_{\text{new}} R^{c\top} J_{\text{new}}^\top = J R^c \Sigma R^{c\top} J^\top$$

Follow-up

Why can three nearby training views provide less useful depth information than three spread-out views?

With a small translation baseline, viewing rays can be nearly parallel and triangulation becomes sensitive to image error. A wider baseline can supply stronger depth constraints, while also changing visibility. More separated views still cannot constrain surfaces they never observe.

Test it in the lab

Keep one Gaussian, note its projected axes, and move it farther along the viewing ray. Then change camera yaw. In this lab yaw orbits the camera centre around the scene; it is not pure rotation at a fixed optical centre.

> *Multiple images help when their geometry adds independent constraints, not simply because there are more pixels.*

5.1 · EXPERIMENT DESIGN · GENERALIZATION

Interactive experiment · Source code

A fit from one selected camera has low training MSE but a much larger reserved-view MSE. You want to test whether limited view coverage is contributing to the failure.

Which comparison provides the most relevant controlled evidence?

> **Answer A.** Fit one view and three spread-out views from the same seeds, Gaussian budget, shape mode, and update budget; evaluate both at the reserved pose.

The controlled comparison varies view coverage while holding the initialization and model family fixed. If the additional views improve reserved-view error, that supports the claim that they supply useful constraints for this fixture. Compare both training and reserved behavior rather than swapping the metric between runs.

This is evidence, not a universal proof of geometry recovery. Additional views also change the nonlinear objective, occlusions, and convergence behavior. Equal update counts here average over each selected set; the multi-view run performs more rendering work per update. A computation-matched study would control that too.

$$L_{\text{train}}(\theta) = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \|R(\theta, P_v) - I_v\|^2$$

Follow-up

If you repeatedly choose budgets and stop times using frame 10, is it still an untouched final test?

No. It remains excluded from gradients, but it has become a validation view used for model selection. An unbiased final performance estimate needs additional untouched views. This tutorial uses frame 10 as a diagnostic reserved view, not as a benchmark claim.

Test it in the lab

Restart with one view, 24 Gaussians, and 100 updates; record training and reserved errors. Repeat with three spread-out views and the same settings. Inspect the reserved image and residual structure as well as the two scalar scores.

> *Design comparisons to separate view constraints, model capacity, optimization, and evaluation leakage.*

5.2 · DENSITY CONTROL · MODEL CHANGES

Interactive experiment · Source code

Consider one isolated coloured splat with opacity o. Replace it by two coincident splats with identical shape and colour, each with opacity o′ = 1 − √(1−o). Let φ be their shared projected Gaussian value at a pixel.

Does this opacity adjustment preserve the rendered image everywhere?

> **Answer C.** No: it matches alpha at φ = 1, but the composited falloff generally differs elsewhere.

At the projected centre, φ = 1 and 1 − (1−o′)² = o by construction. Away from the centre, the combined alpha is 2o′φ − o′²φ². That quadratic dependence on φ is generally different from the original oφ. The trivial zero-opacity case is unchanged.

Density refinement in the lab also offsets means and may reduce scales, so it changes the forward model further. More kernels create additional fitting capacity; a heuristic initialization does not preserve the current image or guarantee an immediate decrease in error. Subsequent optimization must make the new capacity useful.

$$\alpha_{\text{old}} = o\phi, \quad o' = 1 - \sqrt{1-o}$$
$$\alpha_{\text{pair}} = 1 - (1 - o'\phi)^2 = 2o'\phi - o'^2\phi^2$$
$$o = 0.75,\ \phi = 0.5: \quad \alpha_{\text{old}} = 0.375, \quad \alpha_{\text{pair}} = 0.4375$$

Follow-up

For an additive scalar field, would two coincident copies with amplitudes a/2 preserve the function?

Yes, if their geometry stays identical: (a/2)φ + (a/2)φ = aφ exactly. This distinction is why arguments about additive basis expansion cannot be transferred unchanged to alpha-composited image fitting.

Test it in the lab

Fit the primitive scene for at least 25 updates. Record error and Gaussian count, apply one density refinement, and inspect the immediate change. Fit another 25 updates and distinguish the representation change from the benefit of subsequent learning.

> *Adding capacity and finding a better prediction are separate operations.*

6.1 · VOLUME RENDERING · OPTICAL THICKNESS

Interactive experiment · Source code

A ray crosses a homogeneous slab of length L = 1 with density σ = 2 and constant RGB. You approximate it with N equal intervals of length δ = L/N, using $\alpha_i = 1 - \exp(-\sigma\delta)$.

What happens to total transmittance if you change N from 10 to 100?

> **Answer B.** It stays exp(−2); each smaller interval has proportionally less optical thickness.

Each interval transmits exp(−σL/N). Multiplying N such factors gives exp(−σL), independent of N. Here T_end = exp(−2), approximately 0.1353. With constant colour, the slab contribution is (1−exp(−2))c, plus the remaining background.

Density has units of inverse distance; alpha is dimensionless and depends on interval length. Holding alpha fixed while increasing N would change the modeled optical thickness. For a spatially varying field, additional samples can improve the numerical integral when they resolve its variation; the homogeneous example is exactly integrated by this rule.

$$\delta = L/N, \quad \alpha_i = 1 - e^{-\sigma\delta}$$
$$T_{\text{end}} = \prod_{i=1}^{N} e^{-\sigma L/N} = e^{-\sigma L} = e^{-2} \approx 0.1353$$

Follow-up

If a thin high-density feature lies entirely between the sample points, does a low sampled density prove the ray is empty?

No. Midpoint samples can miss a thin feature. Increasing or redistributing samples addresses integration error of the current field. It does not by itself add observations, learn the missing structure, or guarantee a correct field. Distinguish field approximation from quadrature error.

Test it in the lab

In the NeRF section, inspect the reference field at yaw 90° and the centre ray. Compare 8 and 64 samples, their interval lengths, alpha values, and T_end. This field is nonuniform, so equality is not expected; explain changes through where samples fall and the σδ products.

> *NeRF density is not a per-sample opacity. Sampling and physical optical thickness must be kept distinct.*

6.2 · NEURAL FIELDS · MULTI-VIEW CONSISTENCY

Interactive experiment · Source code

In a classical NeRF-style model, density is σθ(x), while RGB is cθ(x,d). A proposed variant also lets density depend freely on viewing direction: σθ(x,d).

Why is position-only density a useful constraint for reconstructing a static scene?

> **Answer C.** It makes different cameras query the same spatial extinction field, while allowing directional appearance in the RGB branch.

A static spatial density field should supply consistent extinction at the same location across views. Restricting density to position encourages the cameras to explain images using shared structure. Colour can still depend on direction, allowing view-related appearance changes without directly changing the density at that position.

The network weights are adjusted through a differentiable pixel loss. The network is not given a target density at every sample. Direction-dependent density could make independent per-view explanations easier, while position-only density still does not remove all image-based ambiguities. Coverage, initialization, architecture, and optimization continue to matter.

$$\sigma = \sigma_\theta(x), \quad c = c_\theta(x, d), \qquad \theta \leftarrow \theta - \eta \nabla_\theta L_{\text{pixels}}$$

Follow-up

Is increasing the number of samples per ray the NeRF equivalent of adding more Gaussian primitives?

No. More ray samples change where and how accurately a fixed field is evaluated; network parameter count stays fixed. Adding Gaussians changes the explicit scene parameterization. Sampling also changes the numerical gradients used during subsequent fitting, so it can affect learning even though it does not increase neural-field capacity.

Test it in the lab

Fit 50 updates and note the weight count. Change samples per ray and inspect the changed rendering while weights and update count remain fixed. Then continue fitting. Compare training and reserved-view errors, and connect the distinction between sampling and capacity to Gaussian density refinement.

> *Shared scene structure, directional appearance, quadrature, and model capacity are separate design choices.*

7.1 · APPEARANCE · REPRESENTATION CHOICE

Interactive experiment · Source code

Suppose a glossy leaf changes brightness with viewing direction in the captured photographs, but a reconstruction represents each Gaussian with one fixed RGB value. Even with good spatial coverage, a highlight appears painted onto the surface.

Which model change directly addresses this missing behavior?

> **Answer B.** Let each Gaussian's emitted colour depend on viewing direction, for example through spherical harmonics.

The spatial Gaussian specifies local support; the appearance model specifies colour. A fixed RGB value cannot directly express a view-dependent highlight. Direction-dependent colour coefficients add that degree of freedom. Full 3DGS uses spherical harmonics for this purpose; the compact Garden viewer used here displays RGB appearance.

Additional spatial splats may partly compensate for an inadequate appearance model, but that can entangle geometry and colour rather than model the directional effect directly. Higher output resolution changes sampling, not the trained appearance parameters. A richer appearance model also needs observations that constrain it.

$$c_i(d) = \sum_{\ell,m} \beta_{i,\ell m} Y_{\ell m}(d)$$

Follow-up

Could a low photometric error prove that positions and covariances are correct when colour is very flexible?

No. Geometry and appearance can compensate for each other under finite observations. Evaluate new viewing directions and, when geometry matters, use independent geometric evidence. Directional colour increases expressiveness; it is not a uniqueness guarantee.

Test it in the lab

In section 4, isolate one Gaussian and orbit it. Its footprint changes while its assigned RGB remains fixed. In the Garden, look for view-related appearance changes or limitations, but do not infer the model's training loss from the viewer alone.

> *Choose spatial and appearance parameterizations to match the phenomenon being fitted.*

7.2 · TRANSFER · RENDERING VS GEOMETRY

Interactive experiment · Source code

The garden looks sharp from several viewpoints. A colleague wants to treat the visible Gaussian ellipsoids as a reliable surface model for collision detection.

What does the rendering evidence justify?

> **Answer A.** It supports appearance quality from those views; reliable collision geometry needs additional geometric validation and a defined surface representation.

A photometric objective constrains rendered rays. It does not directly supervise a watertight boundary, distances, or normals. Gaussian ellipsoids describe spatial kernels; a 1σ contour is a visualization convention, not a measured surface. Overlapping, partially transparent primitives can explain images without defining unique collision geometry.

For a geometry-dependent application, define how a surface is obtained and validate it against independent depth, geometry, or task-relevant measurements, including previously occluded regions. An accurate renderer can be valuable for novel-view synthesis while remaining insufficient evidence for a different task.

$$\text{Low image error on observed rays} \quad \not\Rightarrow \quad \text{low surface-distance error}$$

Follow-up

If rendering at twice the resolution reveals smaller floaters, has the underlying geometry become worse?

No. Changing output resolution can expose existing artifacts without changing the stored model. Distinguish representation quality, observation coverage, and display sampling when explaining what you see.

Test it in the lab

Explore viewpoints away from the garden's comfortable viewing region. Inspect thin structures, gaps, and floaters. Return to the primitive experiment and compare its reference surfaces with the learned kernels: ask what is measured, what is fitted, and what remains unsupported.

> *Evaluate a learned representation against the intended task, not only the visual plausibility of its renders.*